Last updated: July 5, 2026

JIAYU (RYAN) ZUO

Sophomore Undergraduate
College of Engineering · EE/BME

HomePage · Education Email
+86 18079101014 · Kuni05
Shenzhen, China · Jiayu Zuo

SELF-INTRODUCTION

Hi! I'm a sophomore undergraduate student in the College of Engineering at Southern University of Science and Technology (SUSTech), intending to major in EE/BME. Strong interests in neuroscience and BCI, with goals to explore the mysteries of the brain and develop innovative brain-computer interface technologies.

Currently conducting research at NCC Lab under the supervision of Associate Professor Quanying Liu. For my latest information, please see my personal homepage.

SKILLS

Programming: Python, C, Java, MATLAB, LaTeX

Frameworks: PyTorch, NumPy, MNE

Tools: Git, Docker, Jupyter

Languages: Chinese (Native), English (Fluent, CET-4)

RESEARCH PROJECTS (LEADING)

05/2026 – Present **EEG-Based Visual Decoding in Real-World Natural Scenes** — **NCC Lab, SUSTech**

- Designed a decoding framework targeting single-trial EEG responses to complex natural scene stimuli, eliminating reliance on trial-averaging that limits real-world applicability.
- Proposed a subject-agnostic alignment strategy to achieve robust cross-subject generalization without per-subject fine-tuning.

`Single-Trial Decoding / Cross-Subject Generalization / Natural Scene / EEG-to-Image`

03/2026 – Present **Adolescent Brain Development Assessment via Large-Scale fMRI Foundation Model** — **NCC Lab, SUSTech**

Core Member

- Building a foundation model pre-trained on a large-scale private fMRI dataset of Chinese school-age children and adolescents to capture developmental brain patterns.
- Designing downstream evaluation tasks for cognitive and neurodevelopmental assessment, including age prediction and developmental trajectory modeling.

`fMRI Foundation Model / Brain Development`

09/2025 – Present **EEG-to-Text Decoding via Foundation Model Alignment** — **NCC Lab, SUSTech**

- Developed a brain-to-text decoding pipeline aligning EEG representations with large language model embedding spaces via contrastive and generative objectives.
- Extended decoding to multilingual settings, bridging neural EEG signals and natural language text across languages.
- Funded by the Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation ("Climbing Program" Special Funds).

`EEG-to-Text / Contrastive Learning`

06/2025 – 09/2025 **Multi-View 3D Reconstruction System for EEG Electrode Localization** — **NCC Lab, SUSTech**

- Developed a precise electrode positioning system using multi-view 3D reconstruction techniques
- Implemented source localization algorithms and VGGT for visual reconstruction
- Applied YOLO-based object detection for automated electrode identification
- Filed one patent application (under review)

`3D Gaussian Splatting / YOLO / Source Localization`

EDUCATION

08/2024 - 06/2028 **Bachelor of Engineering** — **Southern University of Science and Technology (SUSTech)**

Intended Major: EE/BME
Completed: Calculus, Linear Algebra, College Physics, Fundamentals of Electronic Circuits
In Progress: Data Structures and Algorithms, Probability & Statistics, Signals and Systems

PUBLICATIONS & UNDER REVIEW

**Equal Contribution*

BHA/DHA *(title abbreviated to avoid blind review issues)*
D. Li*, **J. Zuo***, K. Xie, W. Yang, Y. Liu, C. Wei, & Q. Liu — *Under Review (NeurIPS 2026)*

FlatCLIP *(title abbreviated to avoid blind review issues)*
M. Wang, W. Ye, Z. Ning, **J. Zuo**, J. Xia, H. Wen, & Q. Liu — *Under Review (NeurIPS 2026)*

HONORS & AWARDS

05/2026 **College Students' Innovative Entrepreneurial Training Plan Program** **SUSTech**

- Project: *Opti-Rhythm: A Digital Sleep Modulation System Based on a Physiological-Topology Brain-Body Coupling Foundation Model*.
- Selected with a grant of **CNY 20,000**.

12/2025 **Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation** ("Climbing Program" Special Funds) **SUSTech**

- Project: *Foundation Model-Aligned Chinese Semantic Decoding from Brain Signals*.
- Selected as a top-tier undergraduate project (Top 30%) with a grant of **CN¥20,000**.

11/2025 **Award of Excellence (Poster Presentation)** **Department of Biomedical Engineering, SUSTech**

- Title: *MindPilot: EEG-guided Brain Modulation using Closed-loop Visual Stimulation Optimization*.
- Awarded at the 6th SUSTech BME Research Day.

COMPETITIONS

05/2026 **Finalist (About Top 1%), 2026 Mathematical Contest in Modeling (MCM)** **COMAP**

- **Team Leader & Advisor**. Topic: *"Artemis's Choice: Chariot or Elevator?"*.

12/2025 **Winning Prize, 2025(6th) "Greater Bay Area Cup" Financial Mathematical Modeling Contest** **GDSTA**

- **Team Leader**. Topic: *"Comprehensive Evaluation and Development Analysis of Stablecoins"*.
- GDSTA: Guangdong Association for Science and Technology
- Hosted by Guangdong Society for Industrial and Applied Mathematics & Guangdong-Hong Kong-Macau Center for Applied Mathematics(Sun Yat-sen University).

05/2025 **Honorable Mention (H Award), 2025 Mathematical Contest in Modeling (MCM)** **COMAP**

- **Solo Participant**. Topic: *"Cybersecurity Analysis Based on the Global Cybersecurity Index (GCI)"*.

2023 **National Second Prize, China High School Mathematical Contest in Modeling** **Beijing Normal University**

- **Solo Participant**. Topic: *"Game Theoretic Analysis of Medical Resource Runs During the COVID-19 Pandemic"*.

RESEARCH EXPERIENCE

Omni-Intelligence *Shenzhen*
06/2026 – Present *Supervisors: Dr. Chen Wei & Mr. Dongyang Li*
Research Intern working on human preference-aligned post-training for generative models.

Neural Computing and Control Lab (NCC Lab) *Department of Biomedical Engineering, SUSTech*
09/2024 – Present *Supervisor: Associate Professor Quanying Liu*
Undergraduate Researcher focusing on BCI, CHI, and NeuroAI.